The Open Systems Interconnection (OSI) Model

1. What Is OSI Model?

The **OSI (Open Systems Interconnection) Model** is a **7-layer conceptual framework** used to understand how data travels through a network.

It divides networking into layers so each layer has a specific responsibility.

```None
User Request

↓

Layer 7

↓

Layer 6

↓

Layer 5

↓

Layer 4

↓

Layer 3

↓

Layer 2

↓

Layer 1

↓

Physical Transmission
```

2. Why OSI Model Matters

Used for:

```
None
Networking fundamentals

Troubleshooting

Protocol understanding

System design interviews

Debugging connection issues
```

3. Easy Memory Trick

```
None
Please Do Not Throw Sausage Pizza Away
```

Means:

```
Physical

Data Link

Network

Transport

Session

Presentation

Application
```

(bottom to top)

4. The 7 Layers

Layer	Name	Purpose	Examples
7	Application	User-facing network services	HTTP, FTP, Telnet
6	Presentation	Data format, encryption	JPEG, UTF, ASCII, SSL/TLS
5	Session	Manage sessions/connections	RPC, NetBIOS
4	Transport	Reliable delivery, ports	TCP, UDP

3	Network	Routing, IP addressing	IP, ICMP
2	Data Link	Frames, MAC addresses	Ethernet, Wi-Fi
1	Physical	Raw bits over wire	Fiber, USB, Cable

5. Layer 7 – Application Layer

Closest to users.

Provides services like:

```None
Web browsing

Email

File transfer

Remote login
```

Protocols:

```None
HTTP

HTTPS

FTP

SMTP

DNS
```

Example:

Browser uses HTTP request.

6. Layer 6 – Presentation Layer

Responsible for:

```None
Data formatting

Encryption

Compression

Encoding
```

Examples:

```None
UTF-8

ASCII

JPEG

MPEG

TLS/SSL
```

Example:

Convert text to UTF-8.

7. Layer 5 – Session Layer

Handles:

```None
Sessions

Connection state

Authentication sessions

Dialog control
```

Examples:

```None
RPC

SQL sessions

NetBIOS
```

Example:

Keep login session alive.

8. Layer 4 – Transport Layer

Provides end-to-end communication.

Protocols:

```None
TCP

UDP
```

Responsibilities:

```None
Segmentation

Reliability

Flow control

Retransmission

Ports
```

Example:

```
None
TCP port 443 = HTTPS
```

9. Layer 3 – Network Layer

Handles routing between networks.

Protocols:

```
None
IP

ICMP
```

Responsibilities:

```
None
Logical addressing

Packet routing

Path selection
```

Devices:

```
None
Routers
```

10. Layer 2 – Data Link Layer

Transfers data inside local network.

Responsibilities:

```None
Frames

MAC addressing

Error detection

LAN communication
```

Examples:

```None
Ethernet

Wi-Fi
```

Devices:

```None
Switches
```

11. Layer 1 – Physical Layer

Actual hardware transmission.

Examples:

```None
Fiber cable

Coax cable

USB

Radio signals

Modem

Repeater
```

Data here is:

```None
0s and 1s

Electrical / Optical / Radio signals
```

12. Real Example: Opening a Website

You open:

```None
https://google.com
```

Flow:

```None
Layer 7 → HTTP request

Layer 6 → TLS encryption

Layer 5 → Session maintained

Layer 4 → TCP packets

Layer 3 → IP routing

Layer 2 → Ethernet/Wi-Fi frame

Layer 1 → Signals over cable/wireless
```

13. Why Important in Interviews

If issue:

```None
Website not loading
```

Think layer-wise:

```None
DNS issue? (L7)

TLS issue? (L6)

Port blocked? (L4)

IP route broken? (L3)

Wi-Fi disconnected? (L2)

Cable unplugged? (L1)
```

Strong debugging method.

14. Modern Note

Real internet mainly uses:

```None
TCP/IP Model
```

But OSI is still best for learning.

15. Common Confusions

HTTP Layer?

```None
Layer 7
```

TCP?

```None
Layer 4
```

IP?

```None
Layer 3
```

Ethernet?

```None
Layer 2
```

16. Interview Talking Point

If asked load balancer:

```None
L4 LB = based on IP/port

L7 LB = based on HTTP path/header/cookie
```

Very common system design topic.

17. One-Line Summary

The OSI model divides networking into 7 layers, from physical hardware transmission up to user-facing applications, making networks easier to understand, design, and troubleshoot.